Emerging Issues in Research and Development

The conceptual and empirical base for implementation science

Tracy Bradfield & Scott McConnell
Center for Early Education and Development
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Today's Slides: http://z.umn.edu/ImplementationScience2

This Part of Today's Session

- Broad overview – **The Science** of Implementation Science
  - Durlak and Dupre (2008)
  - Berkel, Mauricio, Schoenfelder & Sandler (2011)
- A specific example – Implementing Parent Management Training, Oregon Model (PMTO) in new places

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Durlak & Dupre: A Call to the Field
http://z.umn.edu/durlakdupre

- To transfer effective programs into real world settings. There are 4 phases of program diffusion:
  - Dissemination
  - Adoption
  - Implementation
  - Sustainability
- Implementation = key focus of their work

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Little Prior Attention to Implementation

- In 1998, Dane & Schneider found that only 39 of 162 published mental health prevention studies documented level of implementation and only 13 examined the degree to which implementation affected outcomes.
- In 1997, Durlak found that only 5% of 1,200 mental and physical health prevention studies published by 1995 provided any data on program implementation.
- In 2003, Dusenberg et al found that only 9 of several hundred drug prevention studies conducted over a 25 year period provided information on the relationship between implementation and outcomes.

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Focus of the Article

- Research Questions:
  - Does implementation affect outcomes?
  - What factors affect implementation?
- Literature Review: focus on prevention and health promotion programs for children and adolescents. 1976 – 2006
- Resulted in 483 studies summarized in five meta-analyses and 59 additional studies assessing the impact of implementation on outcomes.

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Results

- RQ: Does implementation affect outcomes?
  - Nearly 500 studies evaluated in five meta-analyses indicates that the magnitude of mean effect sizes are at least two to three times higher when programs are carefully implemented.
  - Data from 59 additional quantitative studies: higher levels of implementation are associated with better outcomes, particularly when fidelity or dosage is assessed.

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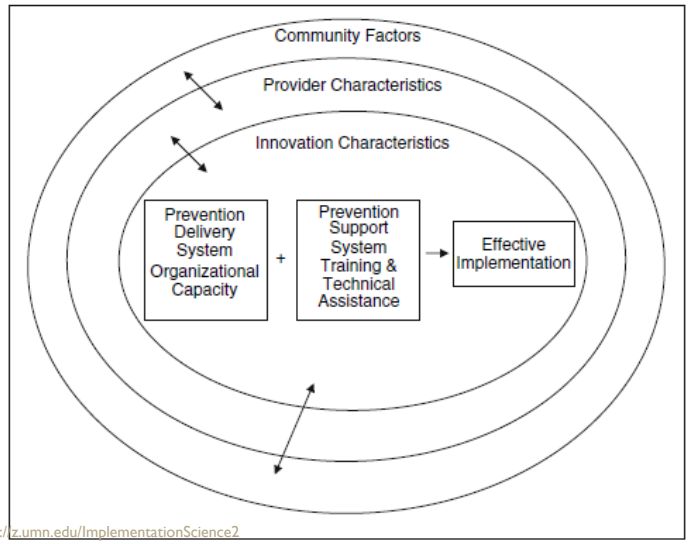
Additional Findings

- Important to include full range of implementation data; don't reduce to "high" vs. "low" which is arbitrary and reduces ability to examine full extent of the relationship between implementation and outcomes.
- Observational data more likely linked to outcomes than self-report

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RQ 2: What factors affect implementation?

Framework for Effective Implementation



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Results

- 81 studies containing quantitative or qualitative data on factors affecting the implementation process.
- Identified 23 factors related to proposed 5 theoretical categories.
- Identified factors were related to implementation in at least five articles with consistent in the more rigorously conducted investigations.

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Factors Affecting the Implementation Process

- Community Level Factors
  - Prevention research system, politics, funding, and policy
- Provider Characteristics
  - Perceived need for innovation, perceived benefit of innovation, self efficacy, and skill proficiency
- Characteristics of the Innovation
  - adaptability and compatibility

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Factors Affecting the Implementation Process

- Organizational Capacity
  - General Organizational Factors
    - Positive Work Climate, Organizational norms, Integration of new programming, Shared vision
  - Specific Practices and Processes
    - Shared decision-making, Coordination with other agencies, Communication, Formulation of tasks (i.e.: workgroups, internal functioning, effective human resource management)
  - Specific Staffing Considerations
    - Leadership, Program Champion, Administrative support

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Factors Affecting the Implementation Process

- Training/ Technical Assistance (Support System)
  - Best Practice: Training that incorporates modeling followed by role playing and performance feedback, specifically when offered in a supportive emotional atmosphere (e.g., Dufrene et al. 2005; Sterling-Turner et al. 2002).
  - Also Important: Early monitoring of implementation followed promptly by retraining. Has been shown to increase the fidelity of implementation to over 85% (DuFrene et al. 2005; Greenwood et al. 2003).

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Next Steps

- Assess the relative influence of different factors on implementation, or the possible interactions that occur among contextual factors.

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Berkel, Mauricio, Schoenfelder & Sandler (2011)

- "Putting the Pieces Together: An Integrated Model of Program Implementation"
- Proposes a theoretical model of relations between the dimensions of implementation and outcomes.

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Aspects of Implementation

(1) Fidelity: the extent to which the innovation corresponds to the originally intended program (sometimes referred to as adherence, compliance, or integrity).

(2) Dosage: how much of the original program has been delivered.

(3) Quality: how well different program components have been conducted.

(4) Participant Responsiveness: the degree to which participants are actively attending.

(5) Program Differentiation: the extent to which a program's theory and practices can be distinguished from other programs.

(6) Program Reach: participation rates and program scope.

(7) Adaptation: changes made in the original program during implementation.

(8) *Monitoring of comparison condition

- This aspect affects outcome evaluation
- Rarely find a true "control" condition in real world field trials

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Four occur within delivery of programs:

- Fidelity (adherence)
- Quality of Delivery (Facilitator Skill in delivery and interaction)
- Participant Responsiveness
- Program Adaptation

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Fidelity

- The extent to which components of the program were delivered as specified.
- Evidence for relation to outcomes: Botvin et al., 1989; Durlak & Dupre, 2008; Forgatch et al., 2005; Ialongo et al., 1999).

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Quality of Delivery

- The process used to convey program material to participants (Dusenbury et al., 2003).
  - Interactive Teaching (Slavin, 1990)
    - Elicits sharing
    - Engages participants in the learning process
  - Facilitator Enthusiasm and Clarity
    - Dane & Scheider, 1998; Hansen et al., 1991
  - Clinical Processes: Reflective Listening & Summarizing
    - Forgatch, 2005
    - Fostering Group Cohesion
      - Coatsworth et al., 2006;

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Quality of Delivery

- In a few studies, quality of process skills has been shown to predict improvements in targeted behaviors (Eames et al., 2009; Forgatch et al., 2005; Giles et al., 2008; Shelef et al., 2005).

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Participant Responsiveness

- Levels of participation and enthusiasm (Dane & Schneider, 1998).
- Operationalized:
  - # of sessions attended
    - Most frequently documented indicator
    - Consistently associated with outcomes (Blake et al., 2001; Kosterman et al., 2001; Prado et al., 2006).
  - Active participation
  - Satisfaction
  - Completion of home practice/ activities.

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Adaptation

- The extent to which facilitators add to or modify content and processes as prescribed in the manual.
- This aspect is a central focus for examining how specific interventions or programs can be tailored to meet the needs of specific cultural groups.
  - i.e.: adaptation to increase the cultural relevance can impact implementation factors such as participant responsiveness and eventual sustainability.
  - Provides link to application of Community Based Participatory Research methodologies

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Four Approaches to Cultural Adaptation

		Knowledge Base	
		Top-down	*Bottom-up*
Effort Orientation	*Individual*	Prevention Research Cycle	Indigenous
	Collaborative	Cultural Adaptation	Culturally Grounded

Lee, Vu, & Lau, in press.

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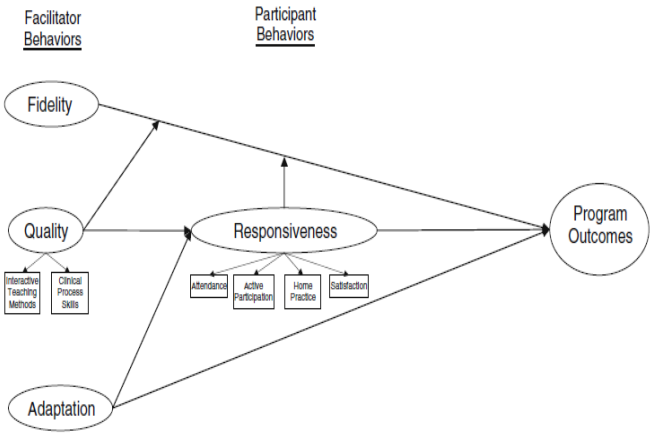
Adaptation

- Durlak and Dupre's review found that all studies that assessed the relation between adaptation and outcomes demonstrated a positive effect.
- Parcel et al., found that adaptation was associated with better outcomes than fidelity.
  - This does NOT mean fidelity is not important. Must first adhere to fidelity in order to understand the core essential components and what can be altered/adapted.

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Proposed Interactional Model of Implementation Factors

Fig. 1 An integrated theoretical model of program implementation



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Proposed Interactional Model of Implementation Factors

- The relation between fidelity and outcomes is mediated by participant responsiveness.
- The link between quality and outcomes is indirect, mediated by increases in participants responsiveness.
  - See next slide
- Quality moderates the relation between fidelity and outcomes.
- Increases in participant responsiveness mediate the link between adaptation and outcomes.

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Relation Between Quality of Delivery and Participant Responsiveness

- Quality is associated with participant responsiveness (Eddy et al., 1998; Rohrbach et al., 2010; Shelef et al., 2005)
  - Facilitator support encourages attendance and active participation.
  - Amount of individual attention facilitators provide predicts attendance (Charlebois et al., 2005).
  - Facilitator skill in promoting group cohesion predicts attendance (Dillman Carpentier et al., 2007).

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Next Steps

- Explicitly testing these pathways
  - Requires adequate measurement of the constructs
    - How best measure responsiveness?
    - How to distinguish fidelity from adaptation?
  - Quantifying behavior to allow for precise measurement may not capture most meaningful components
    - i.e.: attendance much easier capture reliably than ratings of "active participation".
- Test pathways across settings/ delivery contexts
- Outcome specific implementation (related to fidelity)
  - Instead of global implementation measures , focus on specific program components that explicitly target the selected outcomes.

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Implementing Parent Management Training, Oregon Model in New Places

Parent Management Training, Oregon Model (PMTO)

- PMTO
  - evidence-based structured intervention program to help parents and caregivers manage the behavior of the children age 4-12
  - Designed to promote social skills and cooperation and prevent, reduce and reverse the development of moderate to severe conduct problems

Source: http://www.isii.net/website.isii/NewFiles/pmto.html

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Key Features of PMTO

- **Core components**
  - Encouragement of positive behavior
  - Systematic, mild consequences for negative behavior
  - Monitoring
  - Problem-solving
  - Positive involvement
- **Supporting components**
  - Observing and recording behavior
  - Regulating emotions
  - Fostering communication through cooperation
- **Important therapeutic strategies**
  - Supporting and encouraging the development of parenting skills
  - Helping to overcome resistance to change patterns of interaction
  - Using variety in the approach
  - Learning through role-playing and problem solving

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Source: http://www.isii.net/website.isii/NewFiles/pmto.html

For Our Purposes

- PMTO is an evidence-based practice
  - Addresses specific area(s) of development and intervention
  - Specifies necessary conditions for use, training requirements, and manualized intervention procedures
  - Efficacy supported by high-quality research, development, and evaluation
- We can ask "what does it take to implement PMTO in ways that increase the likelihood of efficacy in our setting?

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Replicating PMTO…in Norway

- Odgen, Forgatch, Askeland, Patterson, & Bullock (2005) Implementation of parent management training at the national level: The case of Norway. *Journal of Social Work Practice, 19,* 317-329.
  - Description of adoption, implementation (at "replication" level) of PMTO across Norway, and lessons learned
  - "Top-down" and "bottom-up" initiative

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PMTO Replication/Implementation in Norway

- Conducted across successive cohorts of interventionists
  - Three cohorts reported by 2005
  - Systematic planning and implementation
  - Successive evaluation and adaptation

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Nationwide Implementation Plan

- Establish a national implementation and research center
- Develop plans for regional and local implementation at county and municipal levels
- Establish a comprehensive therapist recruitment, training and maintenance program
- Create a network for collaboration, supervision and quality control
- Conduct clinical outcome research
- Conduct research on the implementation process

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Roll Out

- Therapists
  - 1st Generation – Pioneers, extra skills, recruited as future trainers/ supervisors
    - Agency and individual commitments of time, focus, maintenance of effort
  - 2nd and 3rd Generation – More "mainstream" staff
- Supervisors
  - National/Regional Program Fidelity supervision
  - Local supervisor and peer-supervision
- Experience
  - Local confirmation through empirical evaluation
  - Relevance to local and national perceived need
  - Theoretical conflict of PMTO and therapists
  - Lack of familiarity with implementation of structured intervention
  - Implicit critique of existing services
  - Cultural match, at broad level, between development and implementation contexts

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Why Does This Matter?

- Generally
  - High-fidelity implementation of core components of known-effective intervention increase both confidence of positive effects and, indeed, positive effects
  - Any local limitations or setting-by-intervention interactions will be identified in embedded efficacy evaluation
  - Time to effective intervention and improved outcomes is, presumably, shortened
- Specifically
  - Forgatch, Patterson & DeGarmo (2005) demonstrated that outcomes in replication were affected by the extent that core components of PMTO were implemented and were implemented with high degree of fidelity, was associated with outcomes in replication

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Putting This Part Back Together

- Are there well-developed, well-specified, and ready-to-scale interventions that address one or more issues of importance to you?
- How can adoption with fidelity be facilitated?
  - Top-down
  - Bottom-up
- How will fidelity, efficacy, and need for adaptation be assessed?

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